

Research Question

Our primary research question is: Do medical LLMs exhibit demographic disparities in the frequency and type of hallucinations when generating clinical responses? Prior research has established two important but largely separate findings. On one hand, fairness and health equity studies have shown that LLMs can produce different diagnostic outcomes or recommendations across patient demographics such as race and gender (Rawat et al., 2024; Pfohl et al., 2024; Omar et al., 2025). On the other hand, hallucination research has demonstrated that LLMs frequently generate factual, reasoning, and evidence-related hallucinations in medical contexts, which can pose serious safety risks (Agarwal et al., 2024; Alansari & Luqman, 2026). However, these two streams have not been fully integrated. Fairness evaluations typically rely on multiple-choice accuracy or bias scoring, which do not capture the behavior of open-ended generation. Conversely, hallucination benchmarks usually treat the patient as a demographically neutral entity and do not examine whether hallucination risk differs across groups (Agarwal et al., 2024). This leads to an important open question: if hallucinations occur at different rates for different demographic groups, then the model is not only biased—it may also expose certain populations to systematically higher risks of receiving incorrect or fabricated medical information. Addressing this gap is important because it reframes hallucination not just as a general reliability issue, but as a potential equity and safety issue in medical AI systems (Omar et al., 2025).

Approach

To evaluate this, we will conduct an audit study combining demographic perturbation with a hallucination evaluation framework.

Base Dataset: We will sample high-quality medical queries from the MedHalu dataset (Agarwal et al., 2024), which provides realistic clinical scenarios and a structured hallucination taxonomy.

Demographic Perturbation & Validation: We will systematically inject race and gender identifiers into each query while keeping the clinical content unchanged. To ensure these perturbations do not alter the underlying medical meaning, we will adopt a validation pipeline similar to DiversityMedQA (Rawat et al., 2024).

Generation & Evaluation: For each base query and its demographic variants, we will collect model responses from selected LLMs (e.g., GPT-4o and Llama-3). We will then evaluate each response using an LLM-as-a-judge framework (Zheng et al., 2023), scoring along three hallucination dimensions defined in MedHalu:

- Factual hallucination
- Reasoning hallucination
- Evidence hallucination

Primary Outcome: Our main metric will be the **conditional hallucination rate for each demographic group**, allowing us to compare whether certain groups experience higher rates of hallucinated outputs. We will also analyze hallucination type distributions and intersectional differences (e.g., race × gender).

Sanity Check: To validate the automated evaluation, we will manually annotate a subset (~50–100 responses) and compute agreement between human and model judgments.

Exploratory Analysis: Following recent findings that omissions can co-occur with hallucinations in medical LLM outputs, we will optionally track omission-like failures as a secondary safety signal (Ni et al., 2026).

Significance If successful, this project will map the specific mechanics of how demographic bias manifests in generative medical AI. Instead of just knowing a model is "worse" for minority patients, we will know *how* it fails them—whether it is more likely to invent fake symptoms for Black women or use flawed reasoning for minority men. This granularity is necessary for developing targeted mitigation strategies.

Novelty Our work builds on and connects two existing lines of research, but differs from both in important ways. First, prior fairness work in medical LLMs, such as DiversityMedQA (Rawat et al., 2024) and related health equity evaluation frameworks (Pfohl et al., 2024), focuses primarily on accuracy gaps, recommendation differences, or bias in outputs across demographic groups. While these studies demonstrate that model performance can vary across race and gender, they do not directly measure whether models are more likely to generate factually incorrect or unsupported information for certain groups. Second, hallucination-focused work such as MedHalu (Agarwal et al., 2024) and broader surveys on LLM hallucinations (Alansari & Luqman, 2026) provide detailed taxonomies and evaluation methods for generative errors. However, these benchmarks typically assume demographically neutral inputs and therefore do not address whether hallucination behavior itself varies across populations. Recent work has begun to recognize related concepts such as “bias-induced hallucination” or prompt-dependent hallucination risks (Alansari & Luqman, 2026; Alangari et al., 2025), but these studies do not directly measure demographic differences in hallucination rates under controlled settings. Our project differs by explicitly combining these two directions. By applying a structured hallucination taxonomy to demographically perturbed medical queries, we move beyond both accuracy-based fairness evaluation and demographic-neutral hallucination benchmarks. In doing so, we treat demographic disparity in hallucination as a first-class evaluation target, rather than a side effect of bias or a secondary interpretation of model error.

Feasibility This project is highly feasible within the remaining timeframe given that we leverages existing datasets and proven methodologies:

- **Data:** We already have access to the *MedHalu* dataset, only need to add demographics.
- **Methodology:** The perturbation and validation protocols (from *DiversityMedQA*) and the exact hallucination detection prompts (from *MedHalu*) are documented and reproducible.
- **Evaluation:** By utilizing an automated LLM-as-a-judge backed by a human sanity check (as suggested in the project feedback), we eliminate the traditional bottleneck of requiring large-scale physician annotation, keeping the project well within the scope of a 3-4 week sprint.

Questions:

Another gap we found is that the majority of work only focuses on English context, so we are also considering using different languages to do the same topic. Would you consider this aspect to be more promising and meaningful?

Do we have access to API calls/foundings for this project for closed-weight models?

We plan to test 1 closed-weight model and 1 open-weights model. Given API costs and time constraints, do you have a strong preference or suggestion for which specific open-weights model we should audit for medical use cases?

Related work:

Agarwal, V., Srivastava, A., Garg, S., Sharma, A., Singla, P., & Kaur, M. (2024). MedHalu: Hallucinations in responses to healthcare queries by large language models. arXiv preprint arXiv:2409.19492. <https://arxiv.org/abs/2409.19492>

Crenshaw, K. (1989). Demarginalizing the intersection of race and sex: A Black feminist critique of antidiscrimination doctrine, feminist theory and antiracist politics. *University of Chicago Legal Forum*, 1989(1), 139–167. <https://chicagounbound.uchicago.edu/cgi/viewcontent.cgi?article=1052&context=uclf>

Huang, L., Yu, W., Ma, W., Zhong, W., Feng, Z., Wang, H., Chen, Q., Peng, W., Feng, X., Qin, B., & Liu, T. (2023). A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions. arXiv preprint arXiv:2311.05232. <https://arxiv.org/abs/2311.05232>

Li, J., Cheng, X., Zhao, W. X., Nie, J.-Y., & Wen, J.-R. (2023). HaluEval: A large-scale hallucination evaluation benchmark for large language models. arXiv preprint arXiv:2305.11747. <https://arxiv.org/abs/2305.11747>

Obermeyer, Z., Powers, B., Vogeli, C., & Mullainathan, S. (2019). Dissecting racial bias in an algorithm used to manage the health of populations. *Science*, 366(6464), 447–453. <https://www.science.org/doi/10.1126/science.aax2342>

Pal, A., Umapathi, L. K., & Sankarasubbu, M. (2023). Med-HALT: Medical domain hallucination test for large language models. arXiv preprint arXiv:2307.15343. <https://arxiv.org/abs/2307.15343>

Rawat, R., McBride, H., Ghosh, R., Nirmal, D., Moon, J., Alamuri, D., O'Brien, S., & Zhu, K. (2024). DiversityMedQA: A benchmark for assessing demographic biases in medical diagnosis using large language models. In *Proceedings of the Third Workshop on NLP for Positive Impact (NLP4PI @ EMNLP 2024)*, pp. 334–348. Association for Computational Linguistics. <https://aclanthology.org/2024.nlp4pi-1.29>

Zheng, L., Chiang, W.-L., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., Lin, Z., Li, Z., Li, D., Xing, E. P., Zhang, H., Gonzalez, J. E., & Stoica, I. (2023). Judging LLM-as-a-judge with MT-bench and Chatbot Arena. In *Advances in Neural Information Processing Systems (NeurIPS 2023)*. <https://arxiv.org/abs/2306.05685>

DiversityMedQA: A Benchmark for Assessing Demographic Biases in Medical Diagnosis using Large Language Models. <https://aclanthology.org/2024.nlp4pi-1.29/>
Rawat et al., 2024

Medical Hallucination in Foundation Models and Their Impact on Healthcare. <https://arxiv.org/html/2503.05777v2>
Kim et al., 2025

Large Language Models Hallucination: A Comprehensive Survey. <https://arxiv.org/html/2510.06265v3>
Alansari and Luqman, 2026

A Toolbox for Surfacing Health Equity Harms and Biases in Large Language Models. <https://arxiv.org/html/2403.12025v2>
Pfohl et al., 2024

Disentangling Prompt Element Level Risk Factors for Hallucinations and Omissions in Mental Health LLM Responses. <https://arxiv.org/html/2604.00014v1>
Ni et al., 2026

Evaluating and addressing demographic disparities in medical large language models: a systematic review. <https://link.springer.com/article/10.1186/s12939-025-02419-0>
Omar et al., 2025

Mitigating the risk of health inequity exacerbated by large language models. <https://www.nature.com/articles/s41746-025-01576-4>
Ji et al., 2025